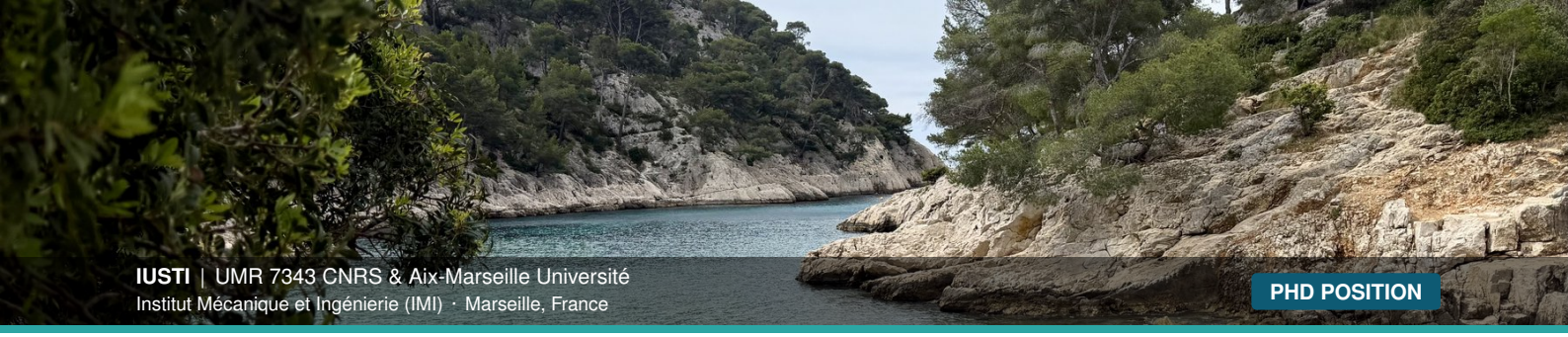


SNSF project *Hierarchy of Scales* · WP 1.3

How does a crack begin in a soft, slender body?

From penny-shaped defects to through-cracks, imaged in 3D

* * *

POSITION	PhD student, experimental	**DURATION**	3 years (doctoral contract)	**START**	As soon as possible
PARTNERS	ETH Zurich (D. Kammer, simulations) · CNRS Lyon (M. Adda-Bedia, theory) · IUSTI & IMI, Aix-Marseille Université				

The sub-project. Most fracture theories describe how an existing crack grows; far less is known about how a crack is born. In soft solids, failure often begins at a small internal defect, such as a cavity or a stiff inclusion, which first opens into a penny-shaped crack, expands in three dimensions, and, once it reaches the free surfaces of a thin specimen, becomes a through-crack [1–4]. This change of topology is controlled by the ratio of the defect size to the sample thickness, and it determines both the load at which a sample fails and the shape of the crack front that emerges [5]. This PhD project (WP 1.3) will develop controlled nucleation experiments in hydrogels: embedding particles of prescribed size and shape in polyacrylamide gels of tunable cross-link density, loading samples of different thicknesses, and imaging the three-dimensional evolution of the nascent crack front by fluorescence and light-sheet microscopy. The student will establish when and how a 3D defect becomes a through-crack, and will compare the measurements with simulations at ETH Zurich and theory in Lyon.

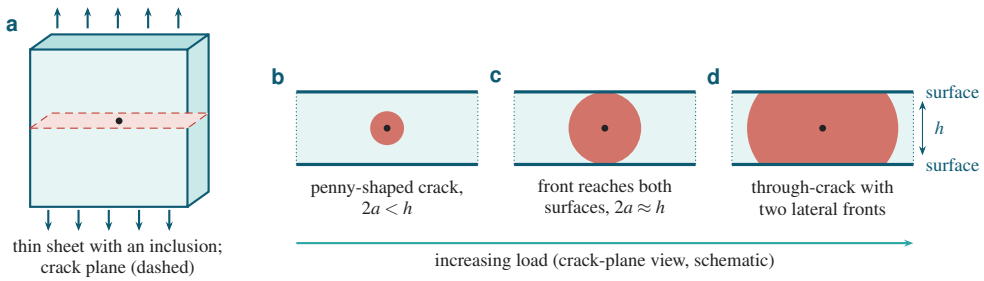


Figure 1. Crack initiation in a slender body (schematic). (a) A thin hydrogel sheet containing a stiff inclusion is loaded in tension. (b–d) Seen in the crack plane, a penny-shaped crack nucleates at the inclusion, grows until its front meets both free surfaces, and transforms into a through-crack. The project will image this 3D-to-2D transition directly and determine how it depends on the defect size a, the sheet thickness h, and the material.

Context. This position is part of the SNSF project "The Hierarchy of Scales in the Fracture of Soft and Slender Bodies" (2025–2029). Whether and how a soft solid breaks depends on several length scales at once: the size of the body itself (its thickness, or slenderness), the nonlinear-elastic zone near the crack tip where the material is stretched far beyond linear response, and the region over which energy is dissipated by molecular processes. In soft materials these scales can be comparable, so classical fracture theories, which rely on their strict separation, break down. The project addresses each scale in a dedicated work package (WP) and connects them through combined experiments, simulations, and theory: experiments at Aix-Marseille Université, simulations in the group of Prof. D. Kammer at ETH Zurich, and analytical theory with Prof. M. Adda-Bedia (CNRS, Lyon). The experimental program is carried out at IUSTI (UMR 7343, CNRS / AMU) in the group of J. Kolinski, equipped with custom tensile loading devices, 3D light-sheet microscopy, and particle-tracking infrastructure, with access to AMU's cryo-electron microscopy facilities. IUSTI and the wider Institut Mécanique et Ingénierie (IMI) have strong expertise in rheology, granular and suspension mechanics, and solid mechanics; collaborations within the lab and across the IMI are strongly encouraged, as are research stays at ETH Zurich. The project will host an international workshop on soft fracture at Monte Verità (Ascona, Switzerland) in November 2027.

Candidate profile. We seek a highly motivated candidate holding, or about to obtain, a Master's degree (M2 or equivalent) in physics, mechanics, materials science, or a related field, with a taste for experiments and quantitative analysis. Experience with optical or fluorescence microscopy, soft materials, or image analysis (Python or MATLAB) is welcome but not required; training will be provided. The PhD will be carried out on a three-year doctoral contract at Aix-Marseille Université, supervised by J. Kolinski, with day-to-day support from the project's postdoctoral researcher and opportunities for research stays at ETH Zurich. The working language of the group is English.

Contact and application. Send a CV, a one-page statement of interest, and the names of two referees to John Kolinski, john.kolinski@univ-amu.fr • kolinski.science. Applications will be reviewed until the position is filled.

[1] A. N. Gent and P. B. Lindley, "Internal rupture of bonded rubber cylinders in tension," *Proc. R. Soc. Lond. A* **249**, 195–205 (1959).
[2] S. Kundu and A. J. Crosby, "Cavitation and fracture behavior of polyacrylamide hydrogels," *Soft Matter* **5**, 3963–3968 (2009).
[3] C. W. Barney *et al.*, "Cavitation in soft matter," *Proc. Natl. Acad. Sci. USA* **117**, 9157–9165 (2020).
[4] S. Raayai-Ardakani, D. R. Earl, and T. Cohen, "The intimate relationship between cavitation and fracture," *Soft Matter* **15**, 4999–5005 (2019).
[5] X. Wei *et al.*, "Complexity of crack front geometry enhances toughness of brittle solids," *Nat. Phys.* **20**, 1009–1014 (2024).